

“Beyond Arduino” Scavenger Hunt

Hints are in the footnotes!

The development board

1. The part number for the development board we’ll be working with is NUCLEO-F303K8. Find the product page for this development board¹. How’s it organized? Which parts of the product page (sections, documents, links, etc) seem useful or important?
2. Try to locate a “getting started” guide, user/reference manual, schematic, or sample projects for the development board. Bookmark or download them; they’ll be useful in helping answer the next few questions.
3. What components are on the development board (e.g. debug adapters, LEDs, buttons, oscillators, etc)? Where are they located and to which pins on the target microcontroller or debug microcontroller are they connected?²
4. Which IDEs support this development board?³
5. What’s the pinout for the development board?⁴

¹ <https://www.st.com/en/evaluation-tools/nucleo-f303k8>

² Check the user manual (in the “Documentation” section of the product page).

³ Check the product page, “getting started” guide, or the user manual (in the “Documentation” section of the product page).

⁴ Check the packaging material or the user manual (in the “Documentation” section of the product page).

6. How can we power the development board?⁵

7. What's the importance of JP1/IDD?⁶

The microcontroller

1. What's the full part number for the microcontroller that's on your development board?
2. Find the product page for the microcontroller⁷. How's it organized? Which parts of the product page (sections, documents, links, etc) seem useful or important? Bookmark or download them for reference later.
3. Locate the circuit diagram for the microcontroller⁸. What features or peripherals does this microcontroller have and what do they mean/do?
4. Try to locate a "getting started" guide, datasheet, reference manual, or sample projects for the microcontroller. Bookmark or download them; they'll be useful in helping answer the next few questions.

⁵ Check the user manual (in the "Documentation" section of the product page), section 6.4. Reading this section in conjunction with the schematics in section 7 helped me figure this out.

⁶ Check the user manual (in the "Documentation" section of the product page), section 6.7. Reading this section in conjunction with the schematics in section 7 helped me figure this out.

⁷ <https://www.st.com/en/microcontrollers-microprocessors/stm32f303k8.html>

⁸ Check the "Circuit Diagram" section of the product page or Table 2 of the datasheet. A more detailed block diagram of the microcontroller can also be found in Figure 1 of the datasheet. The datasheet also has short descriptions for each feature/peripheral.

5. How much memory does the microcontroller have and of what type?⁹
6. What is each pin on the microcontroller used for? Are there any that seem unique or useful? Which peripherals are mapped to each of the microcontroller pins?¹⁰
7. Try to locate the HAL or SDK for this microcontroller and the HAL/SDK reference manual¹¹.
8. What's the supply voltage (Vdd) for the microcontroller (min and max)?¹²
9. Does the microcontroller have any 5V-tolerant pins? Which ones?¹³
10. What's the maximum current that each GPIO pin can source/sink¹⁴?

⁹ Check the "Circuit Diagram" section of the product page or the "Features" and "Ordering" sections of the datasheet.

¹⁰ A partial listing can be found in Table 12 of the NUCLEO-F303K8 user manual. Another great and more detailed summary of the functions available on each pin of the NUCLEO-F303K8 is provided by [Mbed](https://os.mbed.com/platforms/ST-Nucleo-F303K8/) (<https://os.mbed.com/platforms/ST-Nucleo-F303K8/>). The most accurate listing for the STM32F303K8 microcontroller can be found in Tables 13 and 14 of the microcontroller datasheet; you'll need to use one of the pinouts for the NUCLEO-F303K8 to translate the appropriate pin name (e.g. "PA7") to the pin number/location on the development board (e.g. "PA7" corresponds to "A6", the 6th pin on CN4, the right-side header on the development board).

¹¹ The HAL/SDK is called "STM32CubeF3" (under product page > "Tools & Software" > "MCU & MPU Embedded Software") and the reference manual is UM1786, "Description of STM32F3 HAL and low-layer drivers". You shouldn't need to download the HAL; STM32CubeIDE will take care of that for you, though it's useful to know where it exists if you need to go inspect it.

¹² Check Table 19 of the datasheet.

¹³ Check Tables 12 and 13 of the datasheet.

¹⁴ Check Table 17 of the datasheet, $I_{IO(PIN)}$.